



Certificate of Analysis

PFIZER MANUFACTURING BELGIUM NV

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Batch Number: GA2988

Date Generated: 05-May-2022

Product Name: COMIRNATY™ Tris/Sucrose, 30 mcg/0.3 mL Dispersion for Injection (COMIRNATY 0.1mg/ml 195x2.25ml GVL PU BNT)

Material Number: F000054730

Date of Manufacture: 29.03.2022

Expiration Date: 30.11.2022

Importing Country: All countries that accepted Marketing Authorisation Application or WHO Emergency Use Listing

REGISTERED TESTS	ACCEPTANCE CRITERIA	RESULT
COMPOSITION AND STRENGTH		
Appearance (Visual) Appearance	White to off-white suspension	Meets test
Appearance (Particles) Visible Particulates	May contain white to off-white opaque, amorphous particles	Meets test
Subvisible Particulate Matter Subvisible particles	Particles $\geq 10 \mu\text{m}$: ≤ 6000 per container Particles $\geq 25 \mu\text{m}$: ≤ 600 per container	48 Particles $\geq 10 \mu\text{m}$ per container 3 Particles $\geq 25 \mu\text{m}$ per container
Potentiometry pH	7.4 +/- 0.5	7.4
Osmometry Osmolality	320 +/- 80 mOsmol/kg	360 mOsmol/kg
Dynamic Light Scattering (DLS) LNP Size LNP Polydispersity	56 - 101 nm ≤ 0.3	64 nm 0.1
Fluorescence assay RNA Encapsulation RNA Content	$\geq 85\%$ 0.100 +/- 0.026 mg/mL	98 % 0.095 mg/mL
HPLC-CAD ALC-0315 Content ALC-0159 Content DSPC content Cholesterol content	0.90 – 1.85 mg/mL 0.11 – 0.24 mg/mL 0.18 – 0.41 mg/mL 0.36 – 0.78 mg/mL	1.31 mg/mL 0.16 mg/mL 0.28 mg/mL 0.56 mg/mL
Container content Vial content (volume)	Not less than 2.115 mL	≥ 2.115 mL
IDENTITY		
HPLC-CAD Lipid identities	Retention times consistent with references (ALC-0315, ALC-0159, Cholesterol, DSPC)	Meets test

Pfizer Internal Use

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REGISTERED TESTS	ACCEPTANCE CRITERIA	RESULT
RT-PCR Identity of encoded RNA sequence	Identity confirmed	Identity confirmed
POTENCY		
Cell-based Flow Cytometry In Vitro Expression	≥ 30% cells positive	61 %
PURITY		
Capillary Gel Electrophoresis RNA Integrity	≥ 58% intact RNA	66 %
ADVENTITIOUS AGENTS		
Endotoxin (LAL) Bacterial endotoxin	≤ 12.5 EU/mL	<5.00 EU/mL
Sterility Sterility	No growth detected	No growth detected

I HEREBY CERTIFY THAT THE ABOVE INFORMATION IS AUTHENTIC AND ACCURATE.

QUALITY ASSURANCE REVIEW: THE BATCH DOCUMENTATION FOR THE ABOVE LISTED LOT OF PRODUCT HAS BEEN REVIEWED AND ALL ASPECTS WERE FOUND ACCEPTABLE. ALL DEVIATIONS HAVE BEEN THOROUGHLY REVIEWED AND APPROVED. THE RESULTS OF ALL IN-PROCESS TESTING MEET THE REQUIREMENTS. THE BATCH HAS ALSO BEEN TESTED AND CONFORMS TO ALL MAA SPECIFICATIONS AND INTERNAL CONTROL TARGETS. ALL BATCH DOCUMENTATION IS RETAINED AT PFIZER MANUFACTURING BELGIUM NV AND AVAILABLE FOR REVIEW.

MANUFACTURING/PACKAGING REVIEW: THE BATCH DOCUMENTATION FOR THE ABOVE LISTED LOT OF PRODUCT HAS BEEN REVIEWED AND ALL ASPECTS OF THE MANUFACTURING AND PACKAGING WERE JUDGED ACCEPTABLE AND CONSISTENT WITH THE REQUIREMENTS OUTLINED IN THE MAA AND MASTER MANUFACTURING DOCUMENTS. ALL MANUFACTURING DEVIATIONS HAVE BEEN THOROUGHLY REVIEWED AND APPROVED.

ALL ACTIVITIES ARE PERFORMED BY QUALIFIED PEOPLE, UNDER THE SUPERVISION OF THE QUALIFIED PERSON.

Prepared by:

Name: *Aurélie Verniers* Aurélie Verniers
05 May 2022 09:18:051-0400

Title: QP Trainee
Date: 05/05/2022

Signature: REASON: I approve this document.

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Approved by:

Name: *Lise De Pauw* Lise De Pauw
11 May 2022 10:10:031-0400

Title: QP Delegate
Date: 11/05/2022

Signature: REASON: I approve this document.

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BioNTech Manufacturing GmbH

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Manufacturing

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Certificate of Compliance

(No.: CoC- GA2988-01)

Product:	COMIRNATY™
Lot/Batch No.:	GA2988
Date of Manufacture	29 Mar 2022

Batch size:	709 215 vials
Packaging:	GC-Vial 2 mL (195 vials/pack)
Strength/Potency:	30 µg BNT162b2 per dose (0.3 mL)
Volume per vial:	2.25 mL
Dosage Form:	Dispersion for i.m. injection
Retest date/Expiry date:	30 Nov 2022
Destination:	World Health Organisation
Authorization No.:	EU/1/20/1528/003#

Name and Address of Manufacturer (QP release):	BioNTech Manufacturing GmbH Kupferbergterrasse 17 - 19 55116 Mainz, Germany
Manufacturing authorization number /GMP-Certificate:	DE_RP_01_MIA_2021_0046/ DE_RP_01_GMP_2021_0033

Remarks:

WHO release certificate no.BE/22/1162 has been issued by Sciensano, NCL Belgium.
WHO approval dated 23 Nov 2021

Certification Statement:

I hereby certify that all the manufacturing stages of this batch of finished product have been carried out in full compliance with the GMP requirements of the EU and with the requirements of the EU Marketing Authorisation.

11 MAY 2022

Olga Waksman

Date, Signature of the Qualified Person [Olga Waksman]